Supplementary Material

When Simple Is Enough: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction

This document holds the material that supports the manuscript without being part of its argument: the architecture chosen by the inner loop in every outer fold, and the detail of the screening and docking stage, whose results are hypotheses for experimental testing rather than conclusions of the benchmark. Every value is generated by the analysis pipeline and read from the same files as the manuscript.

Benchmark detail

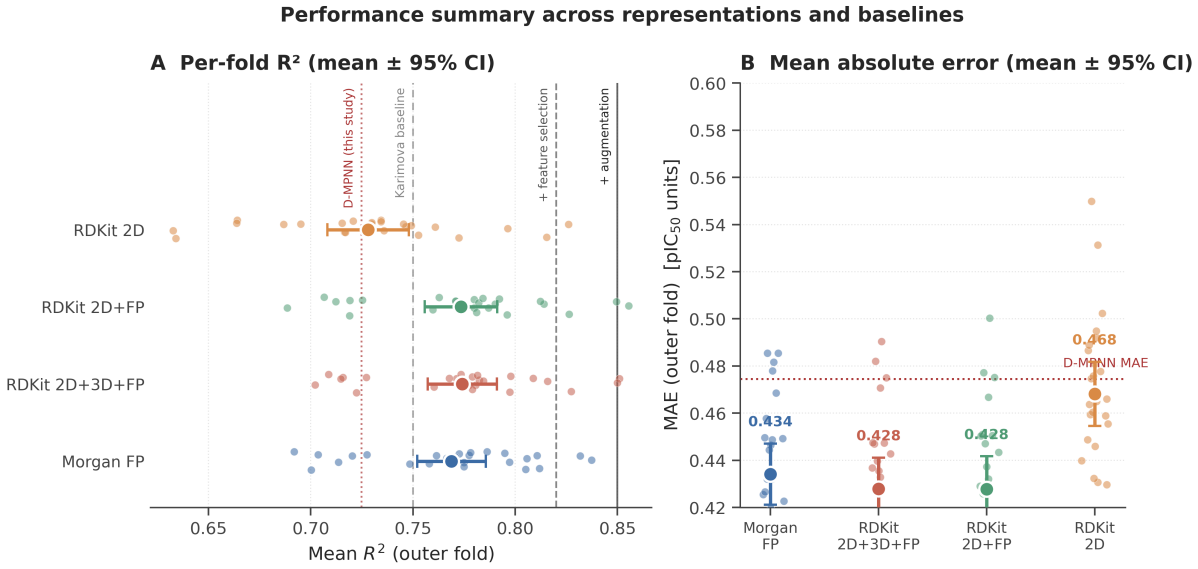


Figure S1: Per-fold performance of the four representations over the 25 outer folds. (A) R^2 of each fold (faint points), with the mean and its 95% confidence interval; (B) the same for the mean absolute error. The vertical lines mark the D-MPNN of this study and the three values reported by the original study, which come from a single hold-out split of a pipeline with the leaks quantified in Section 3.1 of the manuscript and are therefore shown for reference only, not as a paired comparison.

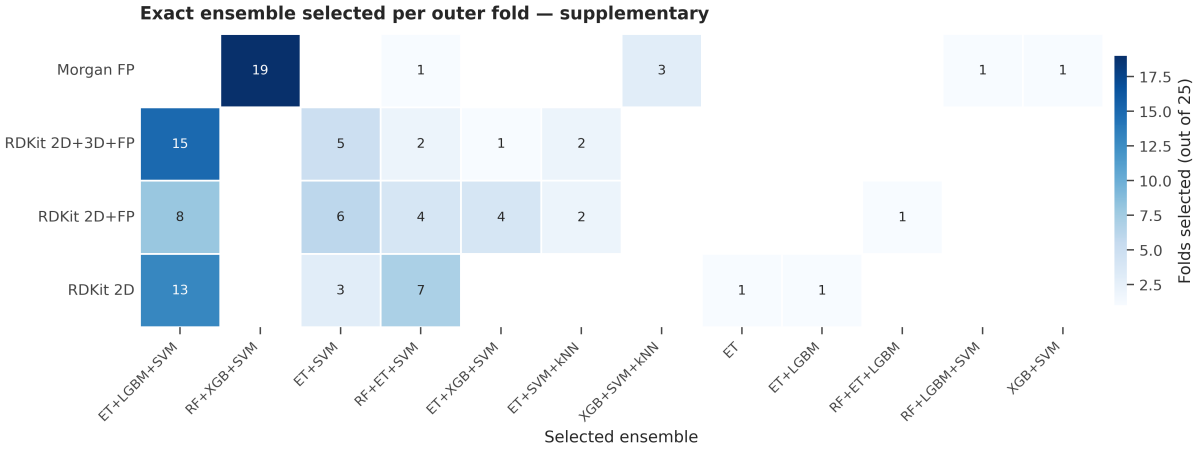


Figure S2: Stacking ensemble selected by the inner loop in each of the 25 outer folds, for each molecular representation. Cell values are the number of outer folds in which that exact combination of base learners (RF, random forest; ET, extra trees; LGBM, LightGBM; XGB, XGBoost; SVM, support vector regression; kNN, k-nearest neighbors) was chosen. No combination was chosen consistently, which shows that the inner search does not converge on a single architecture.

Table S1: Coefficient of determination of every rung of the leakage ladder, seed by seed. R0 is the published protocol, scored on the evaluation partition its own code builds; R1 to R4 remove one leak each and are scored on the 130 molecules common to all of them, and Clean is a reference model with no feature selection. The columns show that the removal of the first three leaks lowers the score in every one of the 10 seeds, while the removal of duplicate structures does not.

Seed	R0	R1	R2	R3	R4	Clean
1	0.848	0.818	0.797	0.709	0.756	0.753
2	0.846	0.820	0.799	0.744	0.751	0.740
3	0.862	0.832	0.788	0.755	0.770	0.759
4	0.849	0.826	0.811	0.774	0.763	0.764
5	0.845	0.825	0.801	0.756	0.706	0.745
6	0.860	0.833	0.782	0.762	0.738	0.757
7	0.862	0.819	0.815	0.747	0.726	0.748
8	0.849	0.826	0.788	0.772	0.707	0.747
9	0.853	0.828	0.823	0.762	0.724	0.765
10	0.859	0.822	0.790	0.739	0.764	0.748
Mean	0.853	0.825	0.800	0.752	0.740	0.752
SD	0.007	0.005	0.013	0.018	0.023	0.008

Table S2: The five variants of the hold-out replication of the original protocol (15% hold-out, seed 42), on a classical stacking ensemble. Bits is the number of fingerprint bits the variant retains and train rows the size of its training matrix after augmentation. The interval on R^2 is a 2000-sample percentile bootstrap of the held-out molecules; the difference compares each variant with the one it modifies, paired on the same resamples. The two leaky variants differ from their clean counterparts only in that the permutation importance is computed on the complete data set, as the original study does.

Variant	**Bits**	**Train rows**	R^2 (95% CI)	ΔR^2 vs reference
A: no selection, no augmentation	2048	730	0.733 [0.627, 0.817]	
B: selection on the development partition	747	730	0.748 [0.647, 0.829]	+0.015 [−0.013, +0.043] *vs A*
C: B plus Gaussian-noise augmentation	747	2190	0.732 [0.623, 0.820]	−0.001 [−0.027, +0.020] *vs A*
B leaky: selection on the complete data set	446	730	0.836 [0.761, 0.892]	+0.088 [+0.047, +0.139] *vs B*
C leaky: B leaky plus augmentation	446	2190	0.826 [0.748, 0.885]	+0.095 [+0.046, +0.153] *vs C*

Table S3: Every pairwise comparison between the four molecular representations over the 25 shared outer folds, in both metrics. Differences are A minus B, the interval is the 90% confidence interval used by the equivalence test, p is the Nadeau–Bengio corrected paired t-test, p_{Holm} its step-down adjustment within the six comparisons of each metric, and p_{equiv} the two one-sided tests against the ± 0.05 margin.

Metric	**Comparison**	**Difference**	**90% CI**	p	p_{Holm}	p_{equiv}
R^2	Morgan vs RDKit 2D	+0.041	[+0.021, +0.061]	0.0020	0.0081	0.2201
R^2	Morgan vs RDKit 2D+FP	−0.005	[−0.016, +0.007]	0.5061	1.0000	<0.0001
R^2	Morgan vs RDKit 2D+3D+FP	−0.005	[−0.016, +0.005]	0.4006	1.0000	<0.0001
R^2	RDKit 2D vs RDKit 2D+FP	−0.045	[−0.063, −0.028]	0.0002	0.0009	0.3288
R^2	RDKit 2D vs RDKit 2D+3D+FP	−0.046	[−0.063, −0.029]	0.0001	0.0008	0.3489
R^2	RDKit 2D+FP vs RDKit 2D+3D+FP	−0.001	[−0.005, +0.004]	0.8243	1.0000	<0.0001
MAE	Morgan vs RDKit 2D	−0.034	[−0.051, −0.017]	0.0025	0.0099	0.0625
MAE	Morgan vs RDKit 2D+FP	+0.006	[−0.003, +0.016]	0.2739	0.7689	<0.0001
MAE	Morgan vs RDKit 2D+3D+FP	+0.006	[−0.003, +0.016]	0.2563	0.7689	<0.0001
MAE	RDKit 2D vs RDKit 2D+FP	+0.040	[+0.026, +0.054]	<0.0001	0.0002	0.1273
MAE	RDKit 2D vs RDKit 2D+3D+FP	+0.040	[+0.028, +0.053]	<0.0001	<0.0001	0.0980
MAE	RDKit 2D+FP vs RDKit 2D+3D+FP	−0.000	[−0.005, +0.005]	0.9664	0.9664	<0.0001

Table S4: The four y-randomization tests of Rücker et al., each with 100 randomizations scored by 5-fold cross-validation. Scope "architecture" randomizes the stacking ensemble on all 2048 bits; scope "procedure" repeats the complete model-building procedure, including the out-of-fold permutation-importance selection that produces the deployed screening model, inside every randomization. The randomization is either a permutation of the response or a replacement of the descriptors by random ones. Z is the distance from the real model to the mean of the randomized ones in units of their standard deviation; p cannot fall below $1/(100+1)$.

Scope	**Randomization**	R^2 (real)	R^2 (randomized)	Z	p
Architecture	Response permuted	0.779	-0.014 ± 0.008	93.9	0.0099
Architecture	Random descriptors	0.779	-0.014 ± 0.008	95.7	0.0099
Procedure	Response permuted	0.772	-0.036 ± 0.017	47.8	0.0099
Procedure	Random descriptors	0.772	-0.096 ± 0.032	27.0	0.0099

Table S5: Composition of the stacking ensembles chosen by the inner loop, per representation. Each percentage is the share of the 25 outer folds in which that base learner entered the selected ensemble; stability is the share taken by the single most frequent exact combination. RF, random forest; ET, extra trees; LGBM, LightGBM; XGB, XGBoost; SVM, support vector regression; kNN, k-nearest neighbors.

Representation	RF	ET	LGBM	XGB	SVM	kNN	**Most frequent**	**Stability**
Morgan FP	84%	4%	4%	92%	100%	12%	RF+XGB+SVM	76.0%
RDKit 2D+3D+FP	8%	100%	60%	4%	100%	8%	ET+LGBM+SVM	60.0%
RDKit 2D+FP	20%	100%	36%	16%	96%	8%	ET+LGBM+SVM	32.0%
RDKit 2D	28%	100%	56%	0%	92%	0%	ET+LGBM+SVM	52.0%

Table S6: Configuration of the D-MPNN selected by the nested search in each outer fold, with the inner root-mean-square error of that configuration and of the default one, and the score the refit model obtained on the held-out fold. It is the counterpart, for the network, of the stacking compositions in Table S5, and shows that the search did not settle on one configuration.

Fold	**Configuration selected**	**Inner RMSE**	**Inner RMSE, default**	R^2	**MAE**
0	hidden 300, depth 5, dropout 0.0	0.603	0.604	0.721	0.473
1	hidden 300, depth 5, dropout 0.2	0.573	0.600	0.736	0.481
2	hidden 300, depth 5, dropout 0.0	0.756	0.759	0.715	0.467
3	hidden 300, depth 5, dropout 0.0	0.688	0.699	0.758	0.448
4	hidden 600, depth 5, dropout 0.0	0.678	0.730	0.756	0.440
5	hidden 300, depth 5, dropout 0.0	0.624	0.647	0.645	0.491
6	hidden 300, depth 5, dropout 0.0	0.599	0.615	0.792	0.412
7	hidden 300, depth 5, dropout 0.0	0.609	0.682	0.745	0.454
8	hidden 600, depth 5, dropout 0.0	0.692	0.748	0.715	0.498
9	hidden 600, depth 5, dropout 0.0	0.690	0.704	0.687	0.504
10	hidden 600, depth 5, dropout 0.2	0.775	0.867	0.730	0.458
11	hidden 300, depth 5, dropout 0.0	0.640	0.693	0.716	0.471
12	hidden 300, depth 5, dropout 0.0	0.531	0.591	0.715	0.471
13	hidden 300, depth 5, dropout 0.0	0.638	0.680	0.721	0.484
14	hidden 300, depth 3, dropout 0.2	0.649	0.677	0.702	0.517
15	hidden 600, depth 3, dropout 0.2	0.598	0.706	0.725	0.479
16	hidden 300, depth 3, dropout 0.2	0.771	0.843	0.808	0.429
17	hidden 300, depth 5, dropout 0.0	0.711	0.726	0.679	0.471
18	hidden 300, depth 5, dropout 0.0	0.617	0.626	0.700	0.499
19	hidden 300, depth 3, dropout 0.0	0.576	0.576	0.711	0.467
20	hidden 600, depth 5, dropout 0.2	0.674	0.689	0.752	0.439
21	hidden 300, depth 5, dropout 0.2	0.545	0.560	0.709	0.476
22	hidden 300, depth 5, dropout 0.0	0.586	0.626	0.753	0.498
23	hidden 300, depth 5, dropout 0.0	0.754	0.810	0.718	0.484
24	hidden 600, depth 5, dropout 0.0	0.684	0.701	0.748	0.447

Table S7: The two D-MPNN variants of the main benchmark against each classical representation, on the 25 shared outer folds. Difference is the mean of the network minus the classical model over the folds, with its Nadeau–Bengio corrected 90% interval; p_{Holm} is adjusted within each metric and p_{equiv} is the two one-sided test against the ± 0.05 margin. The figures quoted in the manuscript are rows of this table.

Metric	Comparison	Difference	90% CI	p_{raw}	p_{Holm}	p_{equiv}
R^2	D-MPNN, default vs Morgan	−0.044	[−0.074, −0.014]	0.0191	0.0477	0.3665
R^2	D-MPNN, default vs RDKit 2D	−0.003	[−0.035, +0.029]	0.8662	0.8662	0.0101
R^2	D-MPNN, default vs RDKit 2D+FP	−0.049	[−0.081, −0.017]	0.0158	0.0477	0.4708
R^2	D-MPNN, default vs RDKit 2D+3D+FP	−0.049	[−0.080, −0.018]	0.0119	0.0477	0.4827
R^2	D-MPNN, tuned vs Morgan	−0.043	[−0.071, −0.015]	0.0159	0.0436	0.3301
R^2	D-MPNN, tuned vs RDKit 2D	−0.002	[−0.037, +0.033]	0.9254	0.9254	0.0129
R^2	D-MPNN, tuned vs RDKit 2D+FP	−0.047	[−0.078, −0.017]	0.0145	0.0436	0.4415
R^2	D-MPNN, tuned vs RDKit 2D+3D+FP	−0.048	[−0.077, −0.019]	0.0102	0.0409	0.4523
MAE	D-MPNN, default vs Morgan	+0.040	[+0.014, +0.067]	0.0157	0.0313	0.2696
MAE	D-MPNN, default vs RDKit 2D	+0.006	[−0.020, +0.032]	0.6800	0.6800	0.0042
MAE	D-MPNN, default vs RDKit 2D+FP	+0.047	[+0.020, +0.074]	0.0067	0.0201	0.4203
MAE	D-MPNN, default vs RDKit 2D+3D+FP	+0.047	[+0.021, +0.072]	0.0048	0.0190	0.4131
MAE	D-MPNN, tuned vs Morgan	+0.036	[+0.011, +0.062]	0.0231	0.0462	0.1816
MAE	D-MPNN, tuned vs RDKit 2D	+0.002	[−0.024, +0.028]	0.8860	0.8860	0.0020
MAE	D-MPNN, tuned vs RDKit 2D+FP	+0.043	[+0.018, +0.068]	0.0074	0.0232	0.3089
MAE	D-MPNN, tuned vs RDKit 2D+3D+FP	+0.043	[+0.018, +0.067]	0.0058	0.0232	0.2993

Table S8: Each additional neural model against each classical representation over the 25 shared outer folds. Differences are the neural model minus the representation, so a negative value in R^2 means the neural model scored lower. p_{Holm} is adjusted within the four representations of each model and metric.

Metric	Comparison	Difference	90% CI	p_{raw}	p_{Holm}	p_{equiv}
R^2	MLP on ECFP4 vs Morgan	−0.060	[−0.104, −0.015]	0.0300	0.0965	0.6458
R^2	MLP on ECFP4 vs RDKit 2D	−0.019	[−0.069, +0.031]	0.5193	0.5193	0.1492
R^2	MLP on ECFP4 vs RDKit 2D+FP	−0.064	[−0.110, −0.019]	0.0241	0.0965	0.7029
R^2	MLP on ECFP4 vs RDKit 2D+3D+FP	−0.065	[−0.111, −0.019]	0.0246	0.0965	0.7080
R^2	Ensemble of 5 D-MPNN vs Morgan	−0.028	[−0.051, −0.005]	0.0505	0.1622	0.0582
R^2	Ensemble of 5 D-MPNN vs RDKit 2D	+0.013	[−0.016, +0.042]	0.4598	0.4598	0.0199
R^2	Ensemble of 5 D-MPNN vs RDKit 2D+FP	−0.033	[−0.059, −0.006]	0.0468	0.1622	0.1365
R^2	Ensemble of 5 D-MPNN vs RDKit 2D+3D+FP	−0.033	[−0.059, −0.007]	0.0406	0.1622	0.1413
R^2	D-MPNN with RDKit 2D descriptors vs Morgan	−0.068	[−0.103, −0.033]	0.0030	0.0121	0.8049
R^2	D-MPNN with RDKit 2D descriptors vs RDKit 2D	−0.027	[−0.068, +0.014]	0.2657	0.2657	0.1764
R^2	D-MPNN with RDKit 2D descriptors vs RDKit 2D+FP	−0.073	[−0.112, −0.034]	0.0040	0.0121	0.8353
R^2	D-MPNN with RDKit 2D descriptors vs RDKit 2D+3D+FP	−0.073	[−0.112, −0.035]	0.0031	0.0121	0.8464
R^2	D-MPNN initialized from CheMeleon vs Morgan	−0.007	[−0.033, +0.018]	0.6262	1.0000	0.0044
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D	+0.033	[−0.002, +0.069]	0.1216	0.4866	0.2164
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D+FP	−0.012	[−0.040, +0.016]	0.4753	1.0000	0.0155
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP	−0.013	[−0.040, +0.015]	0.4357	1.0000	0.0136
MAE	MLP on ECFP4 vs Morgan	+0.065	[+0.032, +0.097]	0.0024	0.0058	0.7770
MAE	MLP on ECFP4 vs RDKit 2D	+0.031	[−0.006, +0.067]	0.1649	0.1649	0.1896
MAE	MLP on ECFP4 vs RDKit 2D+FP	+0.071	[+0.037, +0.106]	0.0017	0.0058	0.8482
MAE	MLP on ECFP4 vs RDKit 2D+3D+FP	+0.071	[+0.037, +0.105]	0.0015	0.0058	0.8517
MAE	Ensemble of 5 D-MPNN vs Morgan	+0.026	[+0.002, +0.050]	0.0775	0.1550	0.0478
MAE	Ensemble of 5 D-MPNN vs RDKit 2D	−0.008	[−0.030, +0.014]	0.5262	0.5262	0.0017
MAE	Ensemble of 5 D-MPNN vs RDKit 2D+FP	+0.032	[+0.008, +0.056]	0.0305	0.0933	0.1083
MAE	Ensemble of 5 D-MPNN vs RDKit 2D+3D+FP	+0.032	[+0.009, +0.055]	0.0233	0.0933	0.0946
MAE	D-MPNN with RDKit 2D descriptors vs Morgan	+0.061	[+0.028, +0.093]	0.0037	0.0073	0.7136
MAE	D-MPNN with RDKit 2D descriptors vs RDKit 2D	+0.027	[−0.002, +0.056]	0.1250	0.1250	0.0906
MAE	D-MPNN with RDKit 2D descriptors vs RDKit 2D+FP	+0.067	[+0.035, +0.100]	0.0018	0.0053	0.8122
MAE	D-MPNN with RDKit 2D descriptors vs RDKit 2D+3D+FP	+0.067	[+0.036, +0.098]	0.0012	0.0048	0.8209
MAE	D-MPNN initialized from CheMeleon vs Morgan	+0.007	[−0.020, +0.035]	0.6431	1.0000	0.0067
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D	−0.027	[−0.055, +0.002]	0.1291	0.5164	0.0886
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D+FP	+0.014	[−0.014, +0.042]	0.3964	1.0000	0.0174
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP	+0.014	[−0.012, +0.040]	0.3729	1.0000	0.0128

Table S9: Comparisons under scaffold-grouped folds: every neural model against every classical representation, and the classical representations against each other. The unadjusted p is given beside the adjusted one because the 90% intervals quoted in the manuscript are not corrected for multiplicity, which is why an interval can exclude zero while the adjusted p is far from significance.

Metric	**Comparison**	**Difference**	**90% CI**	p_{raw}	p_{Holm}	p_{equiv}
R^2	D-MPNN, default vs Morgan	−0.073	[−0.144, −0.002]	0.0919	0.3676	0.7070
R^2	D-MPNN, default vs RDKit 2D	+0.029	[−0.045, +0.103]	0.5067	0.7607	0.3161
R^2	D-MPNN, default vs RDKit 2D+FP	−0.049	[−0.125, +0.026]	0.2737	0.7607	0.4954
R^2	D-MPNN, default vs RDKit 2D+3D+FP	−0.053	[−0.130, +0.024]	0.2536	0.7607	0.5243
R^2	DNN baseline vs Morgan	−0.105	[−0.215, +0.004]	0.1118	0.4470	0.8028
R^2	DNN baseline vs RDKit 2D	−0.003	[−0.114, +0.107]	0.9594	0.9594	0.2380
R^2	DNN baseline vs RDKit 2D+FP	−0.082	[−0.200, +0.036]	0.2453	0.6965	0.6765
R^2	DNN baseline vs RDKit 2D+3D+FP	−0.085	[−0.204, +0.034]	0.2322	0.6965	0.6914
R^2	DNN with selection and augmentation vs Morgan	−0.084	[−0.159, −0.008]	0.0703	0.2812	0.7735
R^2	DNN with selection and augmentation vs RDKit 2D	+0.018	[−0.072, +0.109]	0.7328	0.7328	0.2777
R^2	DNN with selection and augmentation vs RDKit 2D+FP	−0.060	[−0.147, +0.026]	0.2445	0.6514	0.5796
R^2	DNN with selection and augmentation vs RDKit 2D+3D+FP	−0.064	[−0.149, +0.022]	0.2171	0.6514	0.6054
R^2	Ensemble of D-MPNN vs Morgan	−0.039	[−0.106, +0.028]	0.3259	0.9778	0.3919
R^2	Ensemble of D-MPNN vs RDKit 2D	+0.063	[−0.013, +0.139]	0.1684	0.6736	0.6131
R^2	Ensemble of D-MPNN vs RDKit 2D+FP	−0.016	[−0.092, +0.060]	0.7272	1.0000	0.2242
R^2	Ensemble of D-MPNN vs RDKit 2D+3D+FP	−0.019	[−0.096, +0.058]	0.6773	1.0000	0.2493
R^2	D-MPNN initialized from CheMeleon vs Morgan	+0.007	[−0.038, +0.051]	0.8035	0.9247	0.0538
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D	+0.109	[+0.051, +0.166]	0.0035	0.0142	0.9531
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D+FP	+0.030	[−0.019, +0.079]	0.3082	0.9247	0.2471
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP	+0.027	[−0.025, +0.078]	0.3851	0.9247	0.2238
MAE	D-MPNN, default vs Morgan	+0.049	[−0.002, +0.100]	0.1160	0.4638	0.4820
MAE	D-MPNN, default vs RDKit 2D	−0.026	[−0.081, +0.029]	0.4298	0.8935	0.2332
MAE	D-MPNN, default vs RDKit 2D+FP	+0.033	[−0.023, +0.090]	0.3176	0.8935	0.3096
MAE	D-MPNN, default vs RDKit 2D+3D+FP	+0.035	[−0.021, +0.091]	0.2978	0.8935	0.3228
MAE	DNN baseline vs Morgan	+0.016	[−0.030, +0.061]	0.5565	1.0000	0.1058
MAE	DNN baseline vs RDKit 2D	−0.059	[−0.115, −0.002]	0.0885	0.3541	0.6037
MAE	DNN baseline vs RDKit 2D+FP	+0.001	[−0.056, +0.058]	0.9835	1.0000	0.0763
MAE	DNN baseline vs RDKit 2D+3D+FP	+0.002	[−0.054, +0.058]	0.9523	1.0000	0.0790
MAE	DNN with selection and augmentation vs Morgan	+0.012	[−0.050, +0.075]	0.7378	1.0000	0.1587
MAE	DNN with selection and augmentation vs RDKit 2D	−0.062	[−0.132, +0.007]	0.1383	0.5532	0.6170
MAE	DNN with selection and augmentation vs RDKit 2D+FP	−0.003	[−0.074, +0.068]	0.9485	1.0000	0.1328
MAE	DNN with selection and augmentation vs RDKit 2D+3D+FP	−0.001	[−0.072, +0.069]	0.9729	1.0000	0.1250
MAE	Ensemble of D-MPNN vs Morgan	+0.024	[−0.020, +0.069]	0.3561	1.0000	0.1659
MAE	Ensemble of D-MPNN vs RDKit 2D	−0.050	[−0.106, +0.005]	0.1357	0.5427	0.5037
MAE	Ensemble of D-MPNN vs RDKit 2D+FP	+0.009	[−0.048, +0.067]	0.7866	1.0000	0.1181
MAE	Ensemble of D-MPNN vs RDKit 2D+3D+FP	+0.010	[−0.047, +0.068]	0.7584	1.0000	0.1264
MAE	D-MPNN initialized from CheMeleon vs Morgan	+0.001	[−0.030, +0.032]	0.9775	1.0000	0.0058
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D	−0.074	[−0.109, −0.039]	0.0014	0.0056	0.8742
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D+FP	−0.015	[−0.052, +0.022]	0.5033	1.0000	0.0570
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP	−0.013	[−0.052, +0.026]	0.5632	1.0000	0.0603
R^2	Morgan vs RDKit 2D	+0.1020	—	0.0002	0.0015	—
R^2	Morgan vs RDKit 2D+FP	+0.0235	—	0.3332	0.9997	—
R^2	Morgan vs RDKit 2D+3D+FP	+0.0202	—	0.4151	0.9997	—
R^2	RDKit 2D vs RDKit 2D+FP	−0.0786	—	0.0028	0.0139	—
R^2	RDKit 2D vs RDKit 2D+3D+FP	−0.0819	—	0.0031	0.0139	—
R^2	RDKit 2D+FP vs RDKit 2D+3D+FP	−0.0033	—	0.6539	0.9997	—

Virtual screening and molecular docking

Ligand efficiency metrics

Four efficiency metrics contextualize a predicted potency against the physicochemical load that carries it, which mitigates the risk of prioritizing compounds of high predicted potency and unfavourable properties for development [Hopkins et al., 2004]. Ligand efficiency (LE) is the binding energy per heavy atom count (HAC) [Hopkins et al., 2014]:

$$\mathrm{LE} = \frac{1.37 \times pIC_{50}}{\mathrm{HAC}} \tag{1}$$

The factor of 1.37 converts a dissociation constant into a binding free energy, so substituting pIC_{50} for pK_d assumes the two are interchangeable, which holds only under the Cheng–Prusoff conditions [Cheng and Prusoff, 1973]. The assumption is inherited from the data set, whose potencies are IC_{50} values measured in assays whose substrate concentrations are not reported, and it is one reason none of these metrics was used as a filter. The binding efficiency index (BEI) substitutes the heavy-atom count with the molecular weight (MW) [Abad-Zapatero and Metz, 2005]:

$$\mathrm{BEI} = \frac{pIC_{50}}{\mathrm{MW}} \times 1000 \tag{2}$$

Lipophilic ligand efficiency balances potency against lipophilicity, to mitigate the risk of non-specific binding, as the difference between predicted potency and the calculated octanol–water partition coefficient ($\mathrm{LLE} = pIC_{50} - \log P$, Wildman–Crippen method in RDKit). The surface efficiency index evaluates potency relative to the topological polar surface area (TPSA), as a proxy for membrane permeability ($\mathrm{SEI} = (pIC_{50}/\mathrm{TPSA}) \times 100$). Because composite indices combining logarithmic potency values with linear physical parameters can introduce size-dependent mathematical artifacts [Shultz, 2013], the four metrics were computed from the predicted pIC_{50} and reported alongside it for interpretation; candidates were ranked and selected by predicted potency alone.

Docking protocol

The structure-based check complements the ligand-based predictions by assessing complementarity with the binding site, which mitigates the risk of false positives produced by scaffold memorization. AutoDock Vina v1.2.7 [Das et al., 2026, Trott and Olson, 2010] employs an Iterated Local Search optimizer and a hybrid scoring function, calibrated against PDBbind data, to guide flexible ligands into optimal binding conformations [Eberhardt et al., 2021].

Three-dimensional receptor coordinates were retrieved from the Protein Data Bank (*Toxoplasma gondii* DHFR, PDB ID: 6AOG) [wwPDB consortium, 2019, Berman et al., 2003]. The receptor was prepared as a rigid apo structure: all non-protein records (crystallographic waters, ions, cofactors including the bound NADPH, and the co-crystallized ligand) were removed, retaining both protein chains. Hydrogens were added at pH 7.4 and the structure was converted to the PDBQT format with OpenBabel in rigid mode [Babel, 2011]; keeping the polar hydrogens types the backbone N–H groups as hydrogen-bond donors, which the Vina scoring function distinguishes from acceptors (the partial charges written to the file are not used by this scoring function). While excluding the NADPH cofactor omits a potential source of active-site polarization, this apo-receptor approximation is a common simplification in structure-based virtual screening. The Vina search space was centered on the coordinates of the native CP6 ligand in chain B ($x = 3.689, y = 39.992, z = -62.818$) with a cubic grid box of $20.0 \times 20.0 \times 20.0$ Å and an exhaustiveness parameter of 32, which permitted unrestricted exploration of ligand orientations and rotatable bonds within the site [Trott and Olson, 2010].

Prior to screening the machine learning candidates, the docking protocol was checked by re-docking the co-crystallized CP6 ligand into the TgDHFR active site. Because the PDB file of

the ligand has no bond orders, they were assigned from the known structure of pyrimethamine before the docking input was built, and the starting conformer was generated independently of the crystallographic pose. The top-ranked pose reproduced the crystallographic pose with a symmetry-aware heavy-atom RMSD of 1.11 Å (centroid distance 0.35 Å), below the 2.0 Å criterion adopted for validation. The search box was centered on the crystallographic ligand, so the check confirms that the grid parameters and the docking setup are consistent with the known binding site, but it does not demonstrate blind pose prediction, and it was run on a single ligand.

Following validation, Vina evaluated all 17 candidates of the screening step (the 10 top-ranked compounds plus the reference entries). For each compound the largest fragment was retained to remove counter-ions, hydrogens were added, a single 3D conformer was generated with ETKDGv3 (fixed seed) and minimized with MMFF, and a flexible-ligand PDBQT file was prepared with Meeko, keeping the protonation state of the input structure. Docking used a fixed random seed and returned the 3 best poses. Poses were ranked by the predicted binding free energy (ΔG) in kcal/mol, where lower energies denote stronger predicted binding [Feng et al., 2023]; this is an empirical, rigid-receptor estimate and not a measured thermodynamic quantity. The docking ligand efficiency was computed as the best score per heavy atom. To systematically evaluate the consensus between the ligand-based and structure-based approaches, we established quantitative thresholds. Compounds were classified as validated consensus hits only if they exhibited both high ML-predicted potency ($pIC_{50} > 6.5$) and strong docking affinity ($\Delta G < -8.0$ kcal/mol). Compounds meeting the ML threshold but failing the docking criterion were flagged as potential ML false positives. Conversely, compounds with $pIC_{50} < 6.0$ were labeled validated negatives if $\Delta G > -7.5$ kcal/mol and potential ML false negatives if $\Delta G < -8.0$ kcal/mol; all remaining compounds were classed as neutral. Finally, to quantify the monotonic agreement between the two orthogonal methods without assuming linearity, a Spearman rank correlation analysis was conducted between the continuous pIC_{50} predictions and the Vina docking scores (reported in the manuscript).

Screening results

Table S10: Top 10 FDA-approved candidates predicted by the classical ML model, ranked by predicted pIC_{50} and evaluated across standard ligand efficiency metrics. LE, ligand efficiency; BEI, binding efficiency index; LLE, lipophilic ligand efficiency; SEI, surface efficiency index.

Rank	**Compound Name**	**Predicted pIC_{50}**	**LE**	**BEI**	**LLE**	**SEI**
1	Trimetrexate Glucuronate	7.93	0.40	21.46	5.19	6.74
2	Methotrexate	7.44	0.31	16.38	7.18	3.54
3	Triamterene	7.01	0.51	27.67	6.17	5.41
4	Pralatrexate	7.00	0.27	14.67	6.02	3.38
5	Pyrimethamine	6.57	0.53	26.42	4.05	8.44
6	Folic Acid	6.34	0.27	14.36	6.38	2.97
7	Naldemedine Tosylate	6.09	0.20	10.67	2.61	4.31
8	Bendamustine Hydrochloride	6.08	0.36	16.98	2.82	10.43
9	Levomefolate Calcium	6.07	0.25	13.22	6.33	3.00
10	Letermovir	5.97	0.20	10.42	0.26	7.67

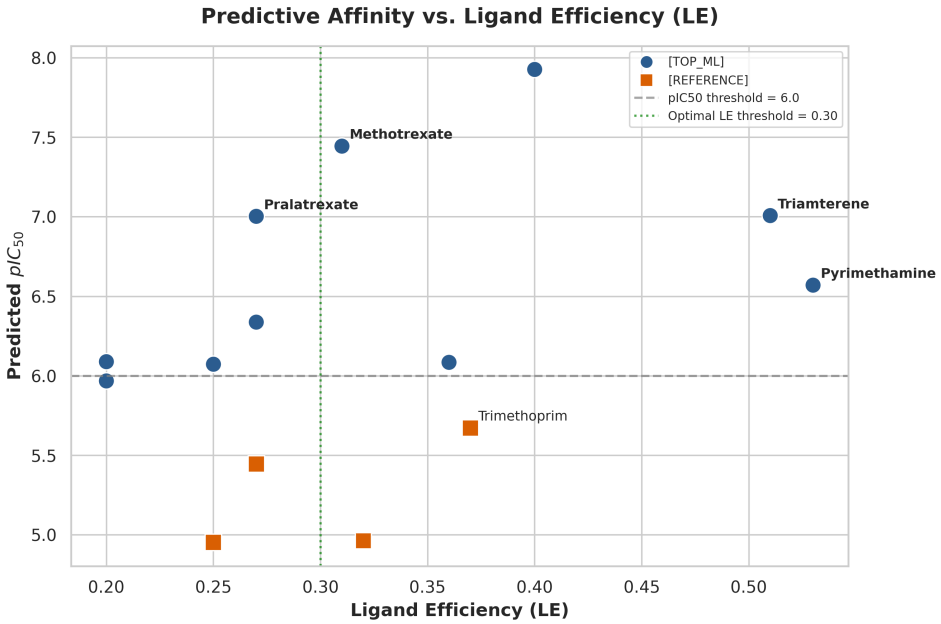


Figure S3: Predicted pIC_{50} versus ligand efficiency for the candidates advanced to docking. Blue: the ten top-ranked compounds by predicted potency; orange: reference compounds from the original study. The dashed line marks $pIC_{50} = 6.0$ and the dotted line LE = 0.30.

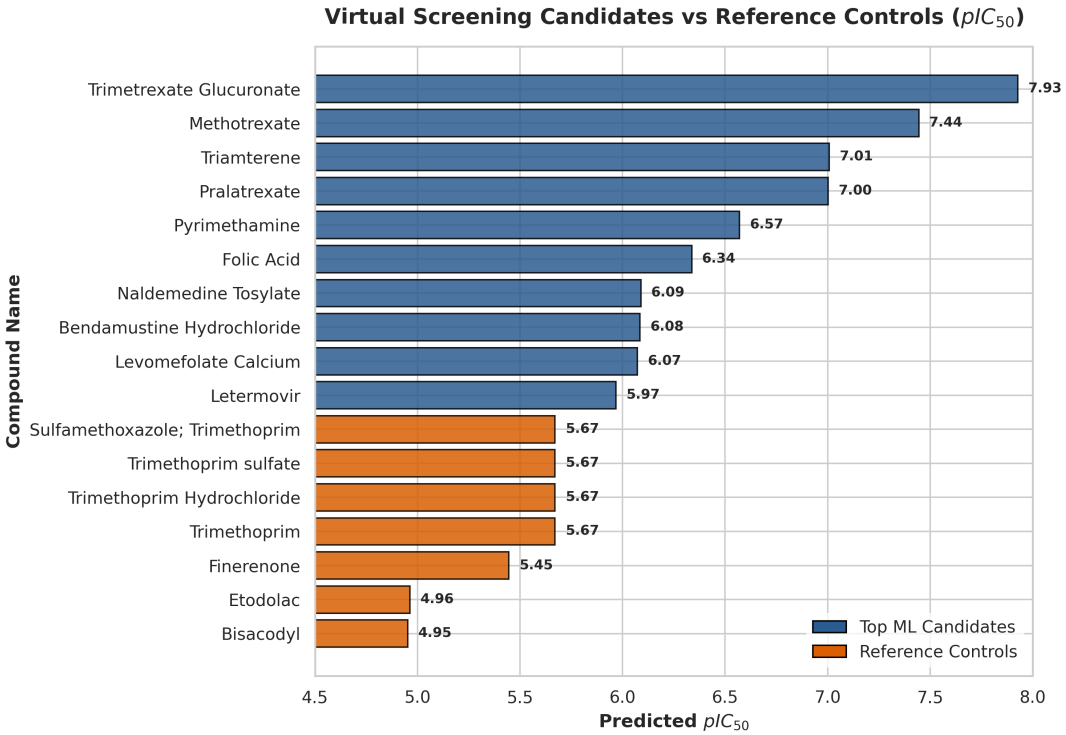


Figure S4: Predicted pIC_{50} of the screening candidates next to the reference controls of the original study. Compounds that belong to the training partition are fitted values and not predictions on unseen molecules.

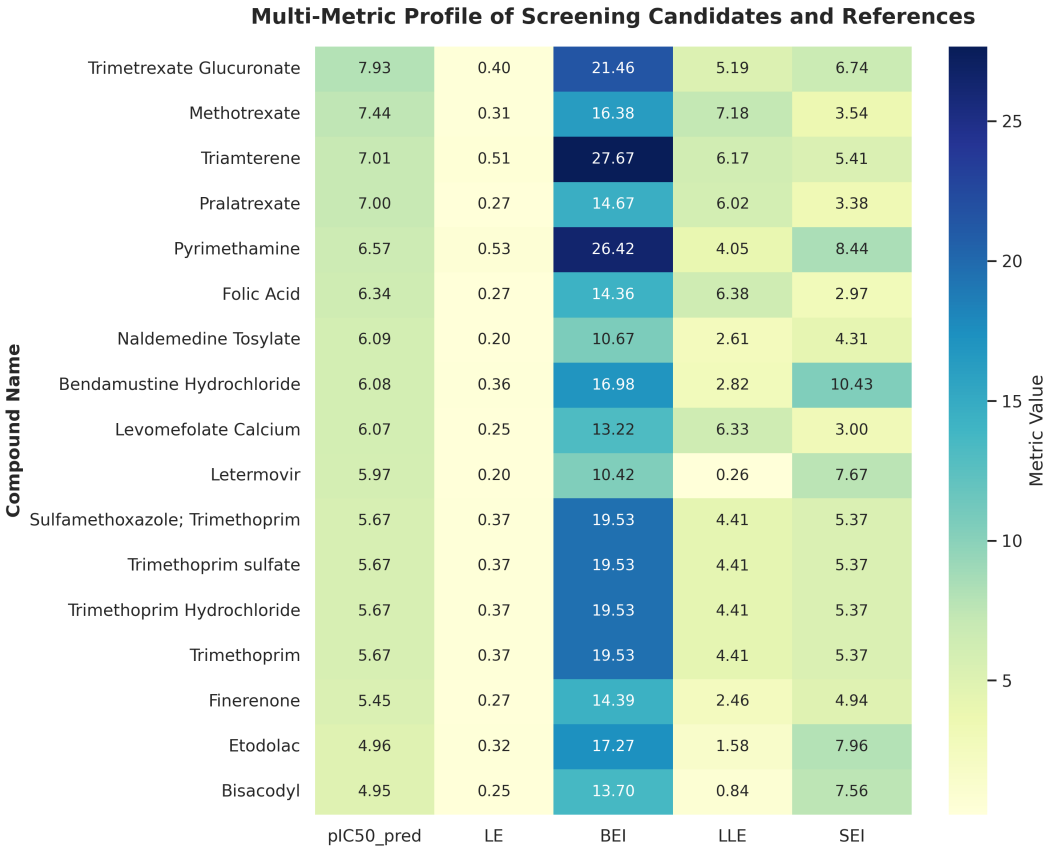


Figure S5: Efficiency metrics of the screening candidates and the reference controls. Each metric is scaled across compounds, so the colour shows a compound's standing within this set and not an absolute quality.

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